

Planetary Phase Dynamics

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**Planetary Phase Dynamics: The Omega Axiom and Numerical
Validation via NASA Data** Jun Ho Ryu February 11, 2026

Abstract

We present the Omega Axiom, a non-Lagrangian, closed-phase framework unifying energy, inertia, gravitation, and electromagnetic phenomena[cite: 5]. By introducing Z-phase, A-wave, and B-effect, the framework links local phase compression to global dynamics[cite: 6]. This paper utilizes the core compression constant $I = 1.50$ to analyze the physical characteristics of Earth and Venus[cite: 86]. By inputting NASA observational data into the defined equations, we derive numerical consistencies regarding energy coefficients and gravitational efficiency[cite: 87]. The analysis demonstrates a 99.3% alignment between the theoretical $I = 1.50$ constant and the observed Venusian atmospheric nodal spacing[cite: 115].

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1 Introduction

Conventional physics treats inertia and gravitation as fundamental forces or geometric necessities[cite: 11]. The Omega Axiom proposes that energy arises from phase compression in closed topologies, and apparent forces are residuals of phase desynchronization[cite: 12]. Electromagnetism is treated as the sole fundamental field, whose closed-phase dynamics yield macroscopic phenomena[cite: 13].

2 The Omega Axiom and Core Equations

2.1 Axiomatic Definition

Energy arises from logarithmic phase compression within a closed topological boundary[cite: 15]. All apparent forces are residuals of boundary phase misalignment[cite: 16].

$$E = m_p \cdot I \cdot r \quad (1)$$

where m_p is phase mass, $I = 1.50$ is the universal compression constant, and r is the closure radius[cite: 20].

2.2 Effective Gravity

Apparent inverse-square attraction emerges from averaged local Z-phase gradients[cite: 55].

$$G_{eff}(r) \propto \frac{\ln I \cdot \langle Z(x, t) \rangle_r}{r^2} \approx \frac{0.405 \cdot \langle Z \rangle}{r^2} \quad (2)$$

3 Phase Dynamics Components

3.1 Z-Phase and Dynamic Lambda

$Z(x, t)$ represents active phase transmission[cite: 43]. The framework integrates this into global FRW expansion[cite: 45]:

$$\left(\frac{\dot{a}(t)}{a(t)} \right)^2 = \frac{8\pi G}{3} \left[\rho_{matter}(t) + \frac{1}{V} \int_V \delta \rho_Z(x, t) d^3x \right] \quad (3)$$

This generates a dynamic Λ without invoking dark energy[cite: 70].

$$\Lambda_{dyn}(t) = 8\pi G \langle \rho_{tot}(x, t) \rangle_\Omega \quad (4)$$

3.2 A-Wave and B-Effect

The A-wave predicts atmospheric standing-wave nodes [cite: 32]: $A(\phi) = \ln \frac{r_{n+1}}{r_n}$ [cite: 29]. The B-effect describes energetic excess at boundaries via a smooth sigmoid function[cite: 38]:

$$B(x, t) = \frac{B_0}{1 + e^{-\alpha(\phi_{closed}(t) - P_{crit})}} \quad (5)$$

4 Numerical Validation: Earth vs. Venus

Inputting NASA observational data into the axiomatic equations yields the following comparative metrics[cite: 96]:

Table 1: NASA Observational Data vs. Omega Axiom Derivations

Parameter	Unit	Earth	Venus	Ratio (E/V)
Mean Radius (r)	km	6,371	6,052	1.0527
Compression Index (I)	const.	1.50	1.50	1.0000
Phase Energy Coeff. ($I \cdot r$)	-	9,556.5	9,078.0	1.0527
Logarithmic Index ($\ln I$)	-	0.405	0.405	1.0000
Gravity Denominator (r^2)	$10^6 km^2$	40.59	36.63	1.1081
Observed Mass (M)	$10^{24} kg$	5.97	4.87	1.2258
Atmospheric Pressure	atm	1	92	0.0108

5 Analysis of Results

5.1 Energy Scaling and Gravitational Efficiency

The total energy scale of Earth is approximately 5.27% higher than that of Venus[cite: 101]. Venus is 10.8% more dense in phase alignment than Earth (2.73×10^{-8} vs 2.46×10^{-8})[cite: 105]. The required Z-phase ratio is[cite: 107]:

$$\frac{\langle Z \rangle_{Earth}}{\langle Z \rangle_{Venus}} = \frac{g_E \cdot r_E^2}{g_V \cdot r_V^2} \approx 1.22 \quad (6)$$

6 Core Structure and Boundary Flux

- **Earth Core** ($r_c \approx 3,485$ km): Existence of a solid inner core indicates a Z-phase transition, suggesting stable energy confinement[cite: 111].
- **Venus Core** ($r_c \approx 3,200$ km): A purely liquid core model provides the basis for B-effect transitioning into 92 atm pressure rather than a magnetic field[cite: 112].

7 Conclusion

The atmospheric layer spacing ratio for Venus (≈ 1.49) demonstrates 99.3% consistency with $I = 1.50$ [cite: 115]. The Omega Axiom replaces additive forces with multiplicative phase logic, unifying inertia and gravitation via closed-phase compression[cite: 70].

References

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